Zachary Miguel Allarey

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Education

University of Florida Gainesville, FL

Bachelor of Science in Computer Science, Minor in Innovation (AI Focused) *Aug 2023 – May 2027*

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Machine Learning, Computer Organization, Programming I & II

Experience

Engineering Intern May 2026 – Aug 2026

KCI Technologies *Greenville, SC*

- Built a **FastAPI** backend that processes PDF plans, runs OCR through **Azure Document Intelligence**, and stores results in PostgreSQL and Azure Blob Storage.
- Combined a **Qwen3-VL** vision model with a rules engine to review engineering sheets, flag issues, and generate review-letter drafts.

LPRC Research Intern Aug 2025 – Apr 2026

Loss Prevention Research Council (LPRC) *Gainesville, FL*

- Built a Google Colab tool with Python and pandas that combines monthly Excel attendance reports and generates summary charts to compare trends across periods.
- Built a web app that connects to Gmail, uses AI to draft replies to incoming emails, and saves the drafts back to the inbox.

AI Search Engine Analyst Intern Oct 2024 – Mar 2025

TELUS Digital *Remote*

- Analyzed and rated AI-generated outputs for relevance, accuracy, and language quality to support improvements in search engine ranking systems.

Projects

AI LoL Draft Assistant | *Python, FastAPI, React, LLM APIs* Aug 2026 – Present

- Building a draft assistant that recommends champions from enemy and allied picks, using matchup, synergy, and win-rate stats collected from high-elo ranked matches via the Riot Games API.
- Designing an LLM explanation layer grounded in team-composition stats (damage split, melee count, healing) computed by an automated per-patch pipeline covering **170+** champions.

Challengers – Tennis Match Predictor | *Python, Scikit-Learn* Jan 2025 – Apr 2025

- Built and evaluated ML classifiers (SVM, Logistic Regression, Random Forest) achieving a 94.6% F1-score.
- Performed feature engineering, model tuning, and system performance analysis on real ATP datasets.

CollegeEats | *React, JavaScript, APIs* May 2025 – Aug 2025

- Designed a calorie-tracking web application powered by the USDA Food Database API.
- Led frontend development using React and Tailwind CSS to create responsive dashboards and authentication flows.

Leadership

External Outreach Officer Jan 2024 – Present

Open Source Club (OSC) *University of Florida*

- Organized outreach events and tabling campaigns, increasing active club membership by 20%.

Core Technical Skills

Programming: Python, Java, C++, JavaScript, TypeScript; object-oriented programming, data structures, algorithms
Machine Learning & Data: Scikit-learn, Pandas, multimodal LLMs, statistical analysis, feature engineering, model evaluation
Systems & Software: FastAPI, Azure (Blob Storage, Document Intelligence), PostgreSQL, Docker, REST APIs, requirements analysis, debugging